

AMENDMENT NO. 2 JULY 1995
TO
IS 1038 : 1983 SPECIFICATION FOR STEEL
DOORS, WINDOWS AND VENTILATORS
(Third Revision)

(Page 3, clause 0.3, line 3) — Substitute 'IS 7452 : 1990' for 'IS : 7452-1982'.

(Page 3, foot-note) — Substitute the following for the existing foot-note:

**Specification for hot rolled steel sections for doors, windows and ventilators (second revision).*

[Page 4, clause 0.4(e)] — Insert the word 'openable' between the words 'high' and 'windows'.

(Page 4, clause 0.6) — Add the following matter at the end of the clause:

'The typical diagrams given in the standard are for guidance only but the dimensions, if given in such typical diagrams, shall be adhered to.'

[Page 12, clause 4.3.1(b), second line] — Insert the words 'horizontally glazed' between the words 'two' and 'fixed-lights'.

(Page 12, clause 5.1.1) — Substitute 'IS 7452 : 1990' for 'IS : 7452-1982'.

[Page 12, clause 5.2.1 (see also Amendment No. 1)] — Substitute the following for the existing clause:

'5.2.1 Coupling section K11B shall be used as mullions for coupling the units side by side and coupling section K12B as transom to couple independent units one above the other. The sections are covered in IS 7452 : 1990*. The coupling bar K11B may be used in case lower window is of fixed type.'

(Page 12, clause 5.2.2) — Delete.

(Page 12, clause 5.2.3) — Renumber the existing clause as '5.2.2'.

(Page 12, clause 5.3) — Substitute 'IS 2835 : 1987' for 'IS : 2835-1977'.

(Page 12, foot-notes with '*' and '†' marks) — Substitute the following for the existing foot-notes:

**Specification for hot rolled steel sections for doors, windows and ventilators (second revision).*

†Specification for flat transparent sheet glass (third revision).

(Page 13, Fig. 4) — Delete.

(Pages 14 to 21, Table 1) — Substitute the following for the existing table:

Price Group 1

TABLE 1 GLASS SIZES (CLEARANCE ALLOWED)
(Clause 5.3)

DESIGNATION	QUANTITY	GLASS SIZE (WIDTH × HEIGHT)	NO. OF GLAZING CLIPS AGAINST EACH PANE
(1)	(2)	(3)	(4)
DOORS (see Fig. 2A)			
Side Hang Type — Horizontal Glazing Bars			
		mm	
6HS20	1	466 × 249	2
	4	466 × 283	2
	1	362 × 283	2
8HS20	1	666 × 249	2
	4	666 × 283	2
	1	362 × 283	2
10HS20	2	407 × 240	2
	9	407 × 283	2
	1	303 × 283	2
12HS20	2	507 × 249	2
	9	507 × 283	2
	1	403 × 283	2
6HS21	1	466 × 249	2
	4	466 × 283	2
	1	362 × 283	2
8HS21	1	666 × 249	2
	4	666 × 283	2
	1	362 × 283	2
10HS21	2	407 × 249	2
	9	407 × 283	2
	1	303 × 283	2
12HS21	2	507 × 249	2
	9	507 × 283	2
	1	403 × 283	2
6NS20	1	466 × 833	4
	1	362 × 283	2
	1	466 × 575	2
8NS20	1	666 × 833	4
	1	562 × 283	2
	1	666 × 575	2
(Continued)			

TABLE 1 GLASS SIZES (CLEARANCE ALLOWED) — <i>Contd</i>			
DESIGNATION	QUANTITY	GLASS SIZES (WIDTH × HEIGHT)	NO. OF GLAZING CLIPS AGAINST EACH PANE
(1)	(2)	(3) mm	(4)
10NS20	1	407 × 283	2
	1	303 × 283	2
	2	407 × 833	4
	2	407 × 575	2
12NS20	1	507 × 283	2
	1	403 × 283	2
	2	507 × 833	4
	2	507 × 575	2
6NS21	1	466 × 833	4
	1	362 × 283	2
	1	466 × 575	2
8NS21	1	666 × 833	4
	1	562 × 283	2
	1	666 × 575	2
10NS21	1	407 × 283	2
	1	303 × 283	2
	2	407 × 833	4
	2	407 × 575	2
12NS21	1	507 × 283	2
	1	403 × 283	2
	2	507 × 833	4
	2	507 × 575	2
WINDOWS (see Fig. 2B)			
Side Hung Type — Horizontal Glazing Bars			
5HS9	1	407 × 273	2
	2	407 × 258	2
6HS9	1	507 × 273	2
	2	507 × 258	2
10HS9	2	425 × 273	2
	4	425 × 258	2
12HS9	2	525 × 273	2
	4	525 × 258	2
15HS9	2	425 × 273	2
	4	425 × 258	2
			(Continued)

TABLE 1 GLASS SIZES (CLEARANCE ALLOWED) – Contd

DESIGNATION	QUANTITY	GLASS SIZES (WIDTH X HEIGHT)	NO. OF GLAZING CLIPS AGAINST EACH PANE
(1)	(2)	(3) mm	(4)
18HS9	3	475 × 273	2
	2	525 × 273	2
	4	525 × 258	2
	3	575 × 273	2
5HS12	2	407 × 277	2
	2	407 × 263	2
6HS12	2	507 × 277	2
	2	507 × 263	2
10HS12	4	425 × 277	2
	4	425 × 263	2
12HS12	4	525 × 277	2
	4	525 × 263	2
13HS12	4	425 × 277	2
	4	425 × 263	2
	4	475 × 277	2
18HS12	4	525 × 277	2
	4	525 × 263	2
	4	575 × 277	2
5HS15	2	407 × 277	2
	2	407 × 263	2
	1	435 × 275	2
6HS15	2	507 × 277	2
	2	507 × 263	2
	1	535 × 275	2
10HS15	4	425 × 277	2
	4	425 × 263	2
	2	454 × 275	2
12HS15	4	527 × 277	2
	4	527 × 263	2
	2	554 × 275	2
15HS15	4	425 × 277	2
	4	425 × 263	2
	2	454 × 275	2
	4	475 × 277	2
	1	475 × 275	2

(Continued)

TABLE 1 GLASS SIZES (CLEARANCE ALLOWED) – Contd			
DESIGNATION	QUANTITY	GLASS SIZES (WIDTH X HEIGHT)	NO. OF GLAZING CLIPS AGAINST EACH PANE
(1)	(2)	(3) mm	(4)
18HS15	4	527 × 277	2
	4	527 × 263	2
	2	554 × 275	2
	4	575 × 277	2
	1	575 × 275	2
Side Hung Type – No Glazing Bars			
5NS9	1	407 × 807	4
6NS9	1	507 × 807	4
10NS9	2	425 × 807	4
12NS9	2	525 × 807	4
15NS9	2	425 × 807	4
	1	475 × 835	4
18NS9	2	525 × 807	4
	1	575 × 807	4
5NS12	1	407 × 1 107	6
6NS12	1	507 × 1 107	6
10NS12	2	425 × 1 107	6
12NS12	2	525 × 1 107	6
15NS12	2	425 × 1 107	6
	1	475 × 1 135	6
18NS12	2	525 × 1 107	6
	1	575 × 1 135	6
5NS15	1	407 × 1 107	6
	1	435 × 275	2
6NS15	1	507 × 1 107	6
	1	535 × 275	2
10NS15	2	425 × 1 107	6
	2	454 × 275	2
12NS15	2	525 × 1 107	6
	2	554 × 275	2
(Continued)			

TABLE 1 GLASS SIZES (CLEARANCE ALLOWED) — Contd
DESIGNATION QUANTITY GLASS SIZES
(WIDTH × HEIGHT) NO. OF GLAZING
CLIPS AGAINST
EACH PANE

(1)	(2)	(3) mm	(4)
15NS15	2	425 × 1 107	6
	1	475 × 1 135	6
	2	454 × 275	2
	1	475 × 275	2
18NS15	2	525 × 1 107	6
	1	575 × 1 135	6
	2	554 × 275	2
	1	575 × 275	2

VENTILATORS (see Fig. 2C)
Top Hung Type — Horizontal Glazing Bars

5HT6	2	407 × 249	2
6HT6	2	507 × 249	2
10HT6	4	449 × 249	2
12HT6	4	549 × 249	2
15HT6	4	463 × 263	2
	2	430 × 249	2
18HT6	4	563 × 263	2
	2	530 × 249	2
5HT9	2	407 × 259	2
	1	435 × 273	2
6HT9	2	507 × 259	2
	1	535 × 273	2

Centre Hung Type — Horizontal Glazing Bars

5HC6	2	360 × 226	2
6HC6	2	460 × 226	2
10HC6	4	426 × 226	2
12HC6	4	526 × 226	2
15HC6	4	464 × 263	2
	2	420 × 226	2
18HC6	4	564 × 263	2
	2	520 × 226	2

(Continued)

TABLE 1 GLASS SIZES (CLEARANCE ALLOWED) — <i>Contd</i>			
Designation	Quantity	GLASS SIZES (WIDTH × HEIGHT)	No. of GLAZING CLIPS AGAINST EACH PANE
(1)	(2)	(3)	(4)
		mm	
Top Hung Type — No Horizontal Glazing Bars			
5NT6	1	407 × 507	2
6NT6	1	507 × 507	2
10NT6	2	449 × 507	2
12NT6	2	549 × 507	2
15NT6	2	463 × 535	2
	1	430 × 507	2
18NT6	2	563 × 535	2
	1	530 × 507	2
5NT9	1	407 × 526	2
	1	435 × 273	2
6NT9	1	507 × 526	2
	1	535 × 273	2
Centre Hung Type — No Horizontal Glazing Bars			
5NC6	1	360 × 460	2
6NC6	1	460 × 460	2
10NC6	2	426 × 460	2
12NC6	2	526 × 460	2
15NC6	2	464 × 536	2
	1	420 × 463	2
18NC6	2	564 × 536	2
	1	520 × 563	2
FIXED-LIGHTS (see Fig. 2D)			
Door Height — Horizontal Glazing Bars			
6HF20	6	535 × 283	2
6HF21	6	535 × 283	2
Door Height — No Glazing Bars			
6NF20	1	535 × 867	4
	1	535 × 283	2
	1	535 × 575	2
6NF21	1	535 × 867	4
	1	535 × 283	2
	1	535 × 575	2
(Continued)			

TABLE 1 GLASS SIZES (CLEARANCE ALLOWED) -- Contd

DESIGNATION	QUANTITY	GLASS SIZES (WIDTH X HEIGHT)	NO. OF GLAZING CLIPS AGAINST EACH PANE
(1)	(2)	(3) mm	(4)
Window Height -- Horizontal Glazing Bars			
5HF9	3	435 x 273	2
6HF9	3	535 x 273	2
10HF9	6	463 x 273	2
12HF9	6	563 x 273	2
15HF9	6	463 x 273	2
	3	491 x 273	2
18HF9	6	563 x 273	2
	3	591 x 273	2
5HF12	4	435 x 277	2
6HF12	4	535 x 277	2
10HF12	8	463 x 277	2
12HF12	8	563 x 277	2
15HF12	8	463 x 277	2
	4	491 x 277	2
18HF12	8	563 x 277	2
	4	591 x 277	2
5HF15	4	435 x 277	2
	1	435 x 291	2
6HF15	4	535 x 277	2
	1	535 x 291	2
10HF15	8	463 x 277	2
	2	463 x 291	2
12HF15	8	563 x 277	2
	2	563 x 291	2
15HF15	8	463 x 277	2
	4	491 x 277	2
	2	463 x 291	2
	1	491 x 291	2
18HF15	8	563 x 277	2
	4	591 x 277	2
	2	563 x 291	2
	1	591 x 291	2

(Continued)

TABLE 1 GLASS SIZES (CLEARANCE ALLOWED) – Contd			
DESIGNATION	QUANTITY	GLASS SIZES (WIDTH × HEIGHT)	NO. OF GLAZING CLIPS AGAINST EACH PANE
(1)	(2)	(3)	(4)
		mm	
		Window Height – No Glazing Bars	
5NF9	1	435 × 835	4
6NF9	1	535 × 835	4
10NF9	2	463 × 835	4
12NF9	2	563 × 835	4
15NF9	2	463 × 835	4
	1	491 × 835	4
18NF9	2	563 × 835	4
	1	591 × 835	4
5NF12	1	435 × 1 135	6
6NF12	1	535 × 1 135	6
10NF12	2	463 × 1 135	6
12NF12	2	563 × 1 135	6
15NF12	2	463 × 1 135	6
	1	491 × 1 135	6
18NF12	2	563 × 1 135	6
	1	591 × 1 135	6
5NF15	1	435 × 1 135	6
	1	435 × 291	2
6NF15	1	535 × 1 135	6
	1	535 × 291	2
10NF15	2	463 × 1 135	6
	2	463 × 291	2
12NF25	2	563 × 1 135	6
	2	563 × 291	2
15NF15	2	463 × 1 135	6
	1	491 × 1 135	6
	2	463 × 291	2
	1	491 × 291	2
18NF15	2	563 × 1 135	6
	1	591 × 1 135	6
	2	563 × 291	2
	1	591 × 291	2
			(Continued)

TABLE 1 GLASS SIZES (CLEARANCE ALLOWED) — *Concid*

DESIGNATION	QUANTITY	GLASS SIZE (WIDTH × HEIGHT)	NO. OF GLAZING CLIPS AGAINST EACH PANE
(1)	(2)	(3) mm	(4)
Ventilator Height — Horizontal Glazing Bars			
5HF6	2	435 × 263	2
6HF6	2	535 × 263	2
10HF6	4	463 × 263	2
12HF6	4	563 × 263	2
15HF6	4	463 × 263	2
	2	491 × 263	2
18HF6	4	563 × 263	2
	2	591 × 263	2
Ventilator Height — No Horizontal Glazing Bars			
5NF6	1	435 × 535	2
6NF6	1	536 × 535	2
10NF6	2	463 × 535	2
12NF6	2	563 × 535	2
15NF6	2	463 × 535	2
	1	491 × 535	2
18NF6	2	563 × 535	2
	1	591 × 535	2

[Page 22, clause 6.1, last sentence (see also Amendment No. 1)]—
Substitute the following for the existing sentence:

‘The process of welding adopted shall be flash butt welding or can be any other process as agreed to between the supplier and the purchaser which shall fulfil the requirements given in 6.1.1.’

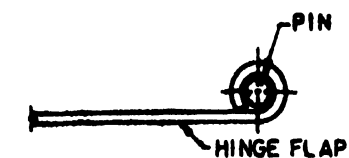
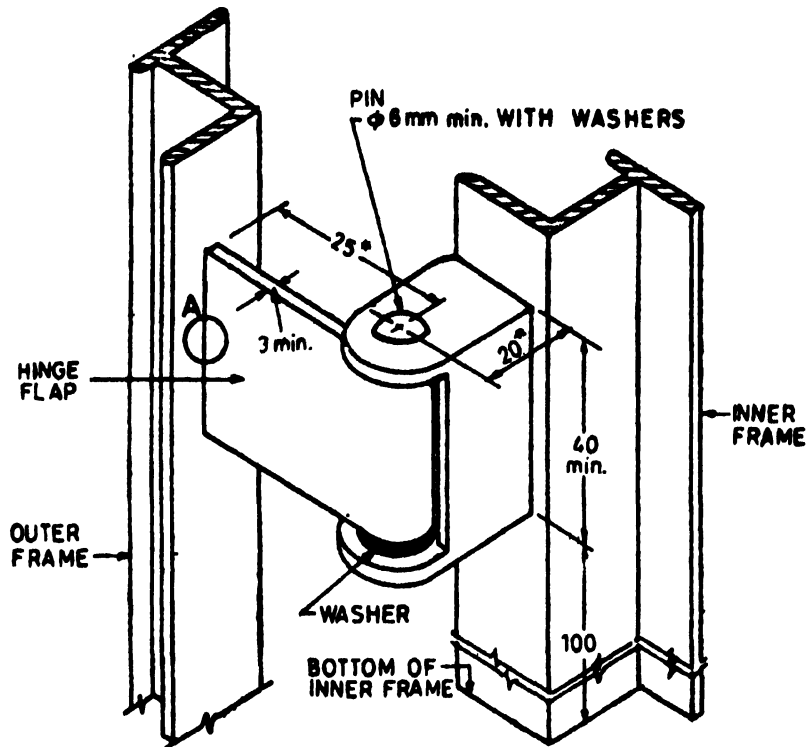
(Page 23, clause 6.1.2) — Substitute ‘tee’ for ‘tie’ in the second line and add the following matter at the end of the clause:

‘All subdividing and glazing bars shall be tennoned and rivetted into the frames.’

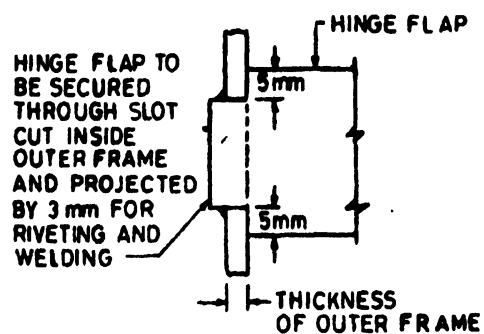
(Page 23, clause 6.2, second para, line 2) — Substitute ‘shall’ for ‘may’.

(Page 23, clause 6.2.1) — Add the following clause after 6.2.1:

‘6.2.1.1 *Non-projecting type of hinges* — Non-projecting type of hinges (see Fig. 14A) may also be used if agreed to between the purchaser and the supplier. The flap shall be of mild steel sheet of thickness not less than 3 mm. The hinge pin of diameter not less than 6 mm and washers shall be of electrogalvanized steel or aluminium alloy.’



**DETAIL OF SECURING
PIN & HINGE FLAP**



DETAIL AT A

NOTES

- 1 The dimensions with '*' mark are to be so adjusted as to ensure close fitting of shutter.
- 2 Hinge box to be welded from inside all through the contact length.
- 3 Hinge flap should take complete round of pin.

All dimensions in millimetres.

**FIG. 14A TYPICAL NON-PROJECTING STEEL HINGE
(Bottom Hinge, Side Hung Shutter)**

(Page 27, Fig. 13) — Substitute 'RIVETTED HINGE PIN' for 'HINGE PIN'.

[Page 28, clause 6.2.2.1 (see also Amendment No. 1)] — Substitute the following for the existing clause:

'6.2.2.1 The handle shall have a two-point nose which shall engage with a brass or aluminium striking plate on the fixed frame in a slightly open position as well as in a fast position (see Fig. 15). Alternatively handle with one point nose may be used, if agreed to between the purchaser and the supplier. The thickness of the handle shall not be less than 3.0 mm for mild steel and brass, and 3.5 mm for aluminium. The height of the handles in each type of side hung shutter shall be fixed in position as indicated in Fig. 16 or as specified by the purchaser.'

(Page 28, Fig. 15):

- a) Delete the matter 'φ 4 mm PIN' and corresponding details for the pin from the drawing.
- b) Substitute '10 mm min OVAL OR SQUARE HOLE' for '11 mm ROUND OR SQUARE HOLE'.

(Page 29, Fig. 16) — Substitute '45 ± 10 cm' for '44.5 cm'.

(Page 29, clause 6.2.3, line 7) — Substitute '1.6 mm min' for '1.25'.

[Page 37, clause 7.1 (see also Amendment No. 1)] — Substitute 'Fig. 29' for 'Fig. 19.'

(Page 41, clause 9.1.1.1) — Add the following matter at the end of the clause:

'Glazing clips shall be used for all sizes of glass panes.'

(Page 41, Notes 1 and 2 under clause 9.1.1.1) — Substitute the following for the existing notes:

NOTE 1 — The holes for glazing clips shall be made during fabrication.

NOTE 2 — In case of doors, windows and ventilators without horizontal glazing bars, the glazing clips may be spaced according to the slots in the vertical member provided the spacing does not exceed 300 mm. The quantity of glazing clips required for each standard size window shall be as given in Table 1.'

(CED 11)